Mathematical Modeling and Secular Variation of Earth's Magnetic Field

David Williams
Independent Study in Mathematics
Montgomery County Community College
Advisor: Dr. Héctor Mera

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Abstract

Earth's magnetic field changes over time. In this project, historical magnetic-pole data from the National Oceanic and Atmospheric Administration (NOAA) were used to build a simple model of Earth's magnetic moment. The data contain north and south magnetic-pole positions for 436 years from 1590 through 2025. Each latitude and longitude pair was changed into Cartesian coordinates on a unit sphere. The vector from the south magnetic pole to the north magnetic pole was then used as a model magnetic moment, $\boldsymbol{m}$. Its magnitude and direction were calculated for every year. The years 1697 and 2025 were studied in detail by hand and then checked with Wolfram Mathematica. From 1697 to 2025, the angles with the x, y, and z axes changed at about $-1.378°$, $0.093°$, and $0.863°$ per century. Over the full 1590–2025 dataset, the modeled north pole traveled about $0.7049R_E$ and the south pole traveled about $0.4755R_E$. Using $R_E = 6371$ km only to give physical meaning, these distances are about 4491 km and 3029 km. The report also explains the centered dipole model, a displaced dipole model, the South Atlantic Anomaly, and why a simple dipole cannot fully describe Earth's real magnetic field.

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1 Introduction

1.1 Wandering of the magnetic poles

The geographic poles are set by Earth's rotation axis, but the magnetic poles move. Their locations change because Earth's magnetic field changes with time. NOAA's National Centers for Environmental Information (NCEI) provides a historical track of the magnetic poles from 1590 through 2025 [1, 2]. This gives a useful way to study *secular variation*, which means slow changes in Earth's main magnetic field over years and centuries.

The main idea of this project is simple. For each year, the magnetic north pole and magnetic south pole are treated as two points on a spherical Earth. The vector from the south pole to the north pole is used as a model for the direction and relative size of Earth's magnetic dipole moment. Repeating the same calculation for every year gives a model that changes with time.

The NOAA pole positions come from geomagnetic field models, not from 436 separate yearly measurements. NOAA explains that magnetic pole positions can be found from the Gauss coefficients used in full magnetic-field models [3]. This project takes that more detailed information and reduces it to a much simpler dipole model.

1.2 Origin of Earth's magnetic field: the geodynamo

Earth's main magnetic field is produced in the liquid outer core. The outer core contains electrically conducting liquid metal. Heat and differences in composition cause this liquid to move. Because the material conducts electricity, this motion can produce electric currents, and those currents produce magnetic fields. This process is called the *geodynamo* [6, 8].

Because the liquid outer core is always moving, the magnetic field is not fixed. The magnetic poles move, the field strength changes, and different parts of the field change shape. The changes studied in this report are one result of that activity inside Earth.

1.3 Magnetic field reversals

Earth's magnetic field has reversed many times in the past. Rocks and sediments can keep a record of the magnetic field that was present when they formed, and these records show past reversals [7]. During a reversal, the large-scale magnetic field changes until the north and south magnetic polarities switch.

The 1590–2025 time range used here is much too short to study a full reversal. Instead, this project studies normal secular variation. Even over a few centuries, the model shows that the field does not behave like a perfectly fixed bar magnet.

1.4 The South Atlantic Anomaly

The South Atlantic Anomaly (SAA) is a large region where Earth's magnetic field is weaker than normal. NASA explains that in this region, charged particles trapped near Earth can come closer to the planet, which can create problems for satellites and spacecraft electronics [9]. NASA also shows that the anomaly changes in position and shape with time [10].

The SAA is important for this project because it shows a clear weakness of a centered dipole model. A perfect centered dipole is very symmetric, so it cannot create a local weak-field region like the SAA. Moving the dipole away from Earth's center can create some asymmetry, but it is still only a simple model. More complete models such as the World Magnetic Model (WMM) and

International Geomagnetic Reference Field (IGRF) use many spherical-harmonic terms instead of only one dipole [4, 5].

2 Data and Mathematical Model

2.1 NOAA magnetic-pole data

The project uses two datasets, `northpole.dat` and `southpole.dat`. Each row has the form

$$(\lambda, \phi, t),$$

where λ is longitude in degrees, ϕ is latitude in degrees, and t is the year. After the files were imported and cleaned, each dataset had 436 rows from 1590 through 2025. The first and last points are listed in Table 1.

Table 1: First and last magnetic-pole locations in the NOAA datasets.

Pole	Year	Longitude (deg)	Latitude (deg)
North	1590	248.169	73.923
South	1590	207.125	-77.964
North	2025	138.057	85.778
South	2025	135.061	-63.851

2.2 Latitude and longitude to Cartesian coordinates

Earth is treated as a sphere with radius $R_E = 1$. Let longitude be λ and latitude be ϕ. The coordinate system used in the hand work and in the notebook is

$$+x : \lambda = 0^\circ, \qquad +y : \lambda = 90^\circ, \qquad +z : \text{North Pole}.$$

The hand derivation starts by separating the radius into a vertical part and a projection in the xy plane. The vertical part is

$$z = R_E \sin\phi,$$

and the size of the projection into the xy plane is

$$\rho = R_E \cos\phi.$$

The x and y parts of that projection are

$$x = \rho\cos\lambda, \qquad y = \rho\sin\lambda.$$

Since $R_E = 1$, the final conversion is

$$\boldsymbol{r}(\lambda, \phi) = \begin{bmatrix} \cos\phi\cos\lambda \\ \cos\phi\sin\lambda \\ \sin\phi \end{bmatrix}. \tag{1}$$

The angles must be changed from degrees to radians before they are used in the trigonometric functions. This derivation was done by hand first and then written as a reusable Wolfram Language function. The full hand work is included in Appendix A.

2.3 Model magnetic moment

For any year, let $\boldsymbol{r}_N$ and $\boldsymbol{r}_S$ be the Cartesian vectors pointing to the north and south magnetic poles. The model magnetic moment is defined as

$$\boldsymbol{m} = \boldsymbol{r}_N - \boldsymbol{r}_S. \tag{2}$$

This is the vector from the south pole to the north pole. Its units are not the real physical units of magnetic moment, so its magnitude is reported in *arbitrary units*. It is still useful because it shows how the geometry of the pole pair changes with time.

The magnitude is

$$\|\boldsymbol{m}\| = \sqrt{m_x^2 + m_y^2 + m_z^2}. \tag{3}$$

The angle between $\boldsymbol{m}$ and each Cartesian axis is found from the dot-product formula. Since the unit vectors $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$, and $\hat{\boldsymbol{k}}$ each have magnitude 1,

$$\alpha_x = \cos^{-1}\left(\frac{m_x}{\|\boldsymbol{m}\|}\right), \tag{4}$$

$$\alpha_y = \cos^{-1}\left(\frac{m_y}{\|\boldsymbol{m}\|}\right), \tag{5}$$

$$\alpha_z = \cos^{-1}\left(\frac{m_z}{\|\boldsymbol{m}\|}\right). \tag{6}$$

These formulas were also worked by hand for 1697 and 2025 before they were used in the computer code.

2.4 Great-circle pole travel

Because the pole positions lie on a sphere, the distance between two nearby yearly positions is measured along the surface. For unit vectors $\boldsymbol{u}$ and $\boldsymbol{v}$, the angle between them is

$$\Delta s = \cos^{-1}(\boldsymbol{u} \cdot \boldsymbol{v}). \tag{7}$$

With $R_E = 1$, this angle in radians is also the arc length in Earth-radius units. Adding the yearly distances gives the total path length. Dividing by the 435-year time span gives an average speed. Multiplying by 6371 km gives an approximate distance in kilometers.

3 Results: 1697 and 2025

3.1 Hand calculation for 1697

For 1697, the pole coordinates were

$$(\lambda_S, \phi_S) = (176.398^\circ, -77.836^\circ)$$

and

$$(\lambda_N, \phi_N) = (247.039^\circ, 74.709^\circ).$$

After converting to radians and using Eq. (1), the hand calculation gave

$$\boldsymbol{r}_{S,1697} \approx \langle -0.210294, 0.013238, -0.977548 \rangle, \tag{8}$$

$$\boldsymbol{r}_{N,1697} \approx \langle -0.102879, -0.242827, 0.964599 \rangle. \tag{9}$$

Therefore,

$$\boldsymbol{m}_{1697} = \boldsymbol{r}_{N,1697} - \boldsymbol{r}_{S,1697} \approx \langle 0.107415, -0.256065, 1.942147 \rangle. \tag{10}$$

The hand calculation also gave

$$\|\boldsymbol{m}_{1697}\| \approx 1.961898,$$

with direction angles

$$\alpha_x \approx 86.8614^\circ, \qquad \alpha_y \approx 97.4996^\circ, \qquad \alpha_z \approx 8.1368^\circ.$$

These values matched the notebook output to rounding.

3.2 Hand calculation for 2025

For 2025, the pole coordinates were

$$(\lambda_S, \phi_S) = (135.061^\circ, -63.851^\circ)$$

and

$$(\lambda_N, \phi_N) = (138.057^\circ, 85.778^\circ).$$

The same coordinate conversion gave

$$\boldsymbol{r}_{S,2025} \approx \langle -0.311959, 0.311295, -0.897651 \rangle, \tag{11}$$

$$\boldsymbol{r}_{N,2025} \approx \langle -0.054760, 0.049208, 0.997286 \rangle. \tag{12}$$

Then

$$\boldsymbol{m}_{2025} \approx \langle 0.257198, -0.262087, 1.894937 \rangle. \tag{13}$$

The magnitude was

$$\|\boldsymbol{m}_{2025}\| \approx 1.930189,$$

and the direction angles were

$$\alpha_x \approx 82.3425^\circ, \qquad \alpha_y \approx 97.8039^\circ, \qquad \alpha_z \approx 10.9670^\circ.$$

Again, the hand work and notebook agreed to rounding. Copies of the original calculations are included in Appendix A.

Table 2: Model magnetic-moment results for 1697 and 2025.

Year	$\|\boldsymbol{m}\|$	α_x (deg)	α_y (deg)	α_z (deg)
1697	1.961898	86.8614	97.4996	8.1368
2025	1.930189	82.3425	97.8039	10.9670

3.3 Rate of angle change

The time from 1697 to 2025 is 328 years. The hand calculation used

$$\text{rate} = \frac{\alpha_{2025} - \alpha_{1697}}{2025 - 1697}(100 \text{ years/century}). \tag{14}$$

This gave

$$\text{Rate}_x \approx -1.3777^\circ/\text{century}, \tag{15}$$

$$\text{Rate}_y \approx \quad 0.0928^\circ/\text{century}, \tag{16}$$

$$\text{Rate}_z \approx \quad 0.8629^\circ/\text{century}. \tag{17}$$

These numbers are average rates between the two selected years. They are not the rate at every point in time. The full-year graphs show that the changes are not perfectly linear.

4 Secular Variation from 1590 to 2025

4.1 Direction of the model magnetic moment

Figure 1 shows the three direction angles for all 436 years. The curves change smoothly, but they are not straight lines. This means there is no single constant rate that describes the whole time period.

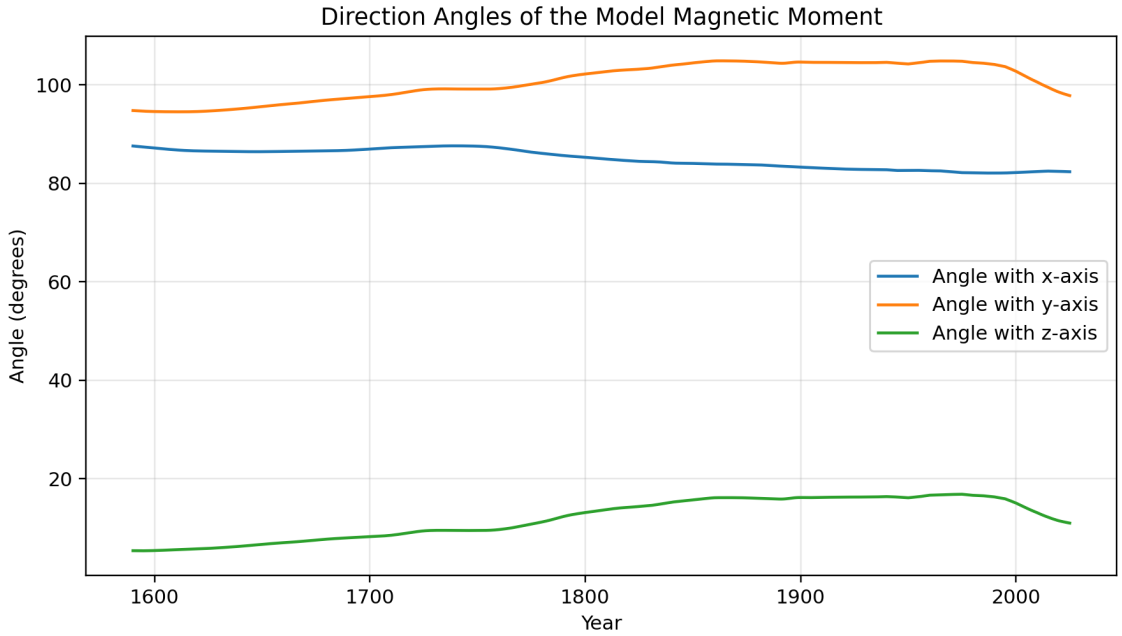


Figure 1: Direction angles of the model magnetic moment from 1590 through 2025.

The x-axis angle falls from about 87.57° in 1590 to 82.34° in 2025. The y-axis and z-axis angles also change by different amounts at different times. This supports the idea that secular variation is a changing process rather than a simple constant rotation.

4.2 Magnitude and Cartesian components

Figure 2 shows the magnitude $\|\boldsymbol{m}\|$. It starts near 1.9474 in 1590, reaches about 1.9698 near 1899, and then falls to about 1.9302 in 2025. Because $\boldsymbol{m}$ is only a geometric model, this graph should not be read as the true physical magnetic moment in ampere-square-meters. It only shows how the pole geometry changes.

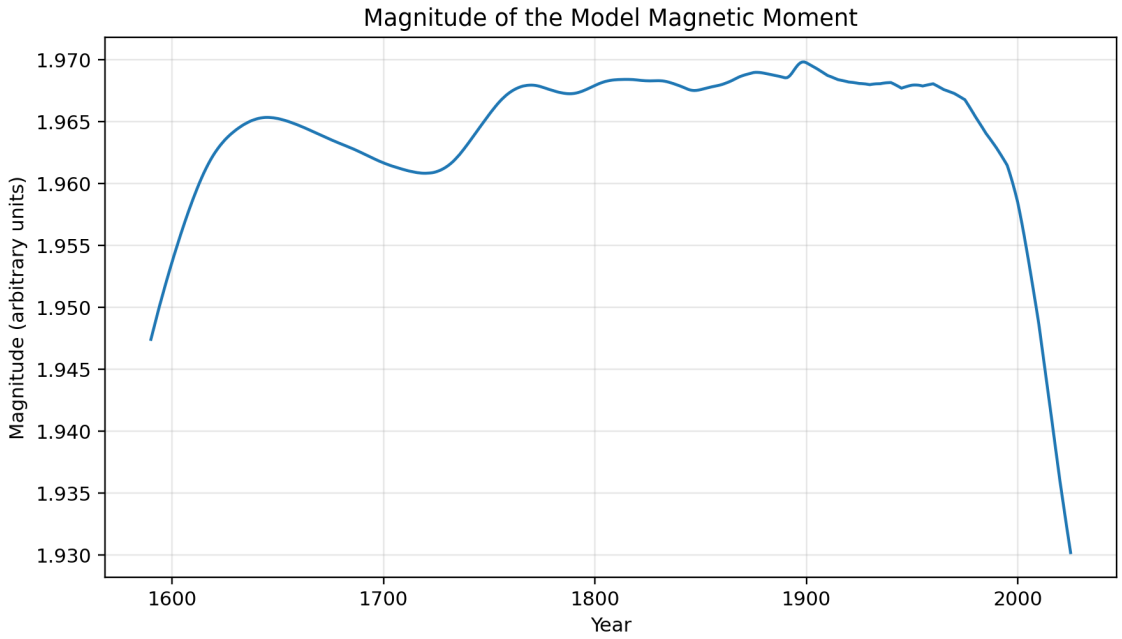


Figure 2: Magnitude of the geometric model magnetic moment. The units are arbitrary.

Figure 3 shows the x, y, and z parts of $\boldsymbol{m}$ separately. The z component stays the largest throughout the dataset, while the x and y parts change more strongly over time.

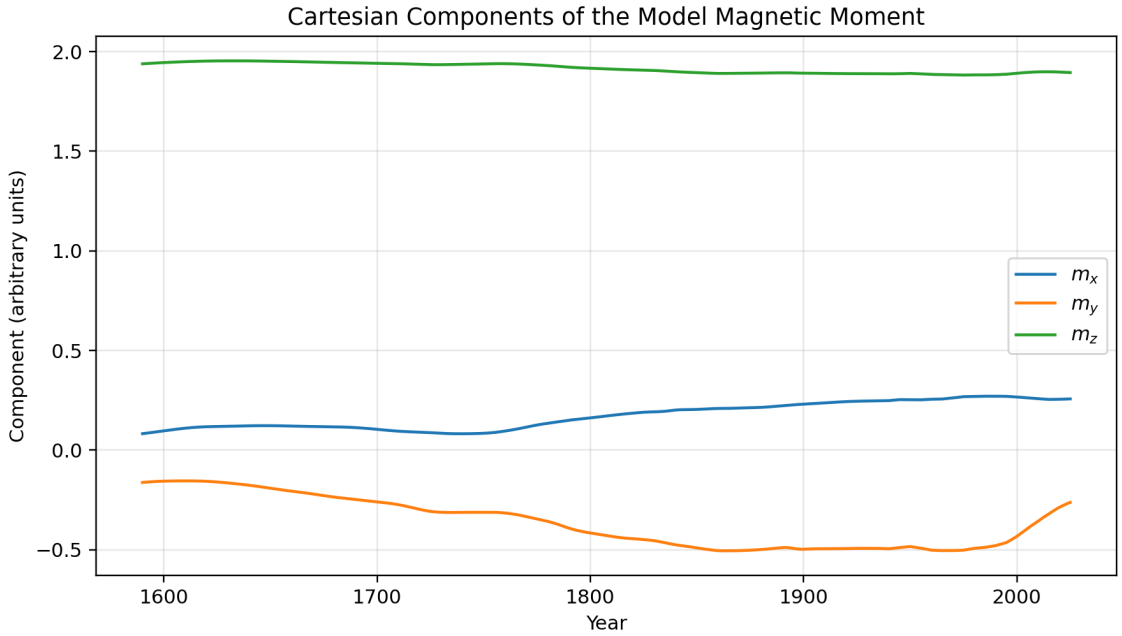


Figure 3: Cartesian components of the model magnetic moment from 1590 through 2025.

4.3 Pole travel and average speed

Adding all yearly great-circle steps gives

$$L_N = 0.7049148 R_E, \tag{18}$$
$$L_S = 0.4754740 R_E. \tag{19}$$

The average speeds over 435 years are

$$\bar{v}_N = 0.00162049 R_E/\text{year}, \tag{20}$$
$$\bar{v}_S = 0.00109304 R_E/\text{year}. \tag{21}$$

Using $R_E = 6371$ km gives

$$L_N \approx 4491.0 \text{ km}, \qquad \bar{v}_N \approx 10.32 \text{ km/year}, \tag{22}$$
$$L_S \approx 3029.2 \text{ km}, \qquad \bar{v}_S \approx 6.96 \text{ km/year}. \tag{23}$$

These are long-term averages. They are not the same as a modern short-term pole speed.

Figure 4 shows the total distance traveled as time passes. The north pole traveled farther in this model, and the changing slope shows that the speed was not constant.

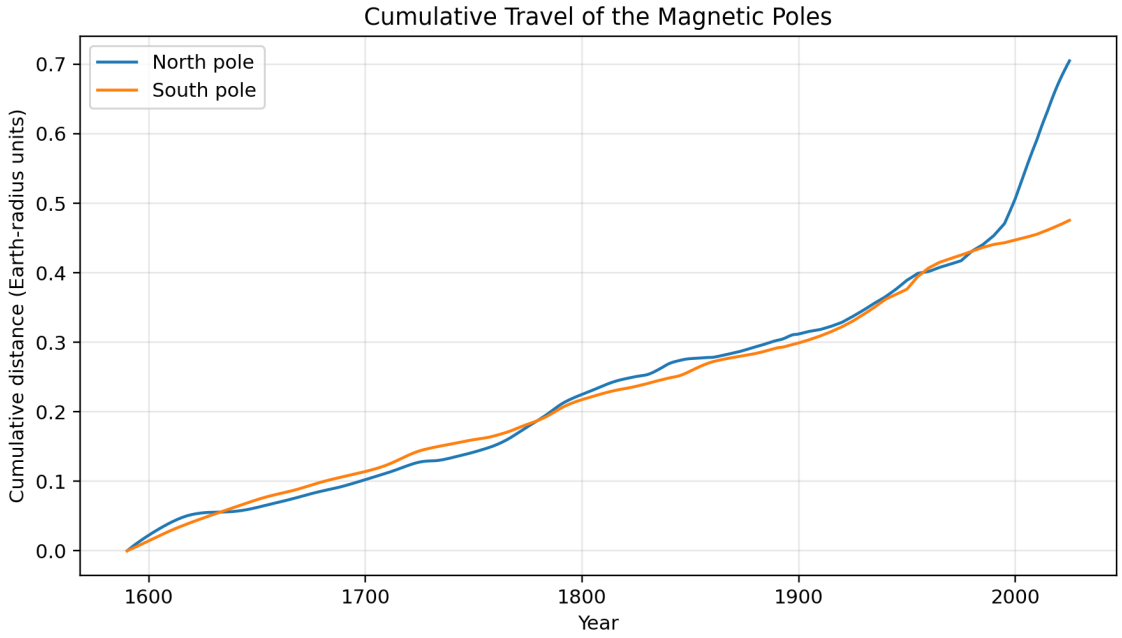


Figure 4: Cumulative great-circle travel of the north and south magnetic poles.

Figure 5 shows the same paths on the unit sphere. The north and south paths are not mirror images. This is one reason to consider a displaced dipole instead of assuming that the field is always a perfect centered dipole.

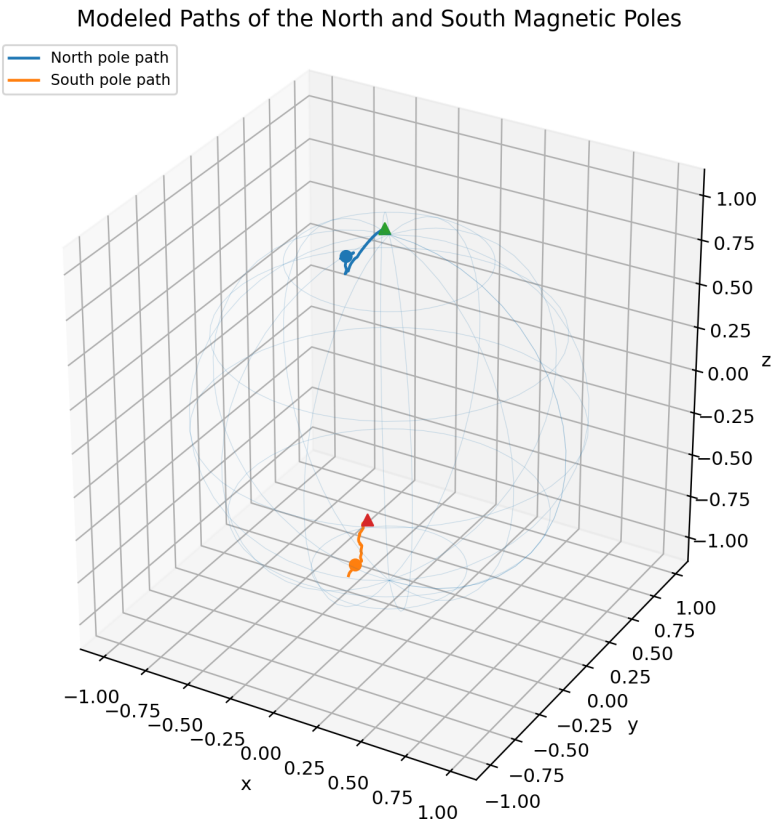


Figure 5: Three-dimensional paths of the north and south magnetic poles on a unit sphere.

5 From a Magnetic Moment to a Magnetic Field

5.1 Centered dipole field

A magnetic dipole moment $\boldsymbol{m}$ at the origin produces the ideal field

$$\boldsymbol{B}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \frac{3(\boldsymbol{m} \cdot \hat{\boldsymbol{r}})\hat{\boldsymbol{r}} - \boldsymbol{m}}{r^3}, \tag{24}$$

where $\hat{\boldsymbol{r}} = \boldsymbol{r}/r$. In the simple version of the model, $\mu_0/(4\pi)$ and R_E are set equal to 1.

At the three unit-sphere Cartesian points, the field becomes

$$\boldsymbol{B}(1,0,0) = \langle 2m_x, -m_y, -m_z \rangle, \tag{25}$$
$$\boldsymbol{B}(0,1,0) = \langle -m_x, 2m_y, -m_z \rangle, \tag{26}$$
$$\boldsymbol{B}(0,0,1) = \langle -m_x, -m_y, 2m_z \rangle. \tag{27}$$

These equations show how the field depends on the direction of $\boldsymbol{m}$.

Along the geographic equator,

$$\boldsymbol{r}(\theta) = \langle \cos\theta, \sin\theta, 0 \rangle, \tag{28}$$

so $r = 1$ and

$$\boldsymbol{m} \cdot \boldsymbol{r} = m_x \cos\theta + m_y \sin\theta.$$

The field magnitude becomes

$$\|\boldsymbol{B}(\theta)\| = \sqrt{\|\boldsymbol{m}\|^2 + 3\left(m_x \cos\theta + m_y \sin\theta\right)^2}. \tag{29}$$

This is useful because it gives a clean formula for a simple centered dipole. It is also very symmetric, which is one of the model's limits.

5.2 Eccentric or displaced dipole

The north and south magnetic poles are not exactly opposite each other. A simple next step is to move the dipole away from Earth's geometric center. One geometric choice is to use the midpoint of the pole vectors,

$$\boldsymbol{c} = \frac{\boldsymbol{r}_N + \boldsymbol{r}_S}{2}. \tag{30}$$

For an observation point $\boldsymbol{r}$, define

$$\boldsymbol{\rho} = \boldsymbol{r} - \boldsymbol{c}. \tag{31}$$

Then the displaced-dipole field is

$$\boldsymbol{B}_{\text{ecc}}(\boldsymbol{r}) = \frac{\mu_0}{4\pi} \frac{3(\boldsymbol{m} \cdot \hat{\boldsymbol{\rho}})\hat{\boldsymbol{\rho}} - \boldsymbol{m}}{\rho^3}. \tag{32}$$

Moving the dipole creates some asymmetry that the centered model does not have. However, it is still only one dipole and cannot match every feature of the real field.

5.3 Why more terms are needed

NOAA's WMM and IGRF are much more complete magnetic-field models [4, 5]. They use spherical-harmonic terms rather than one geometric dipole. In a simple form, the idea can be written as

$$\text{field} = \text{dipole} + \text{quadrupole} + \text{octupole} + \cdots. \tag{33}$$

The dipole is the main large-scale part, which is why the model in this report is useful. The extra terms describe more detail and are needed for a close match to the real field. NOAA also uses many Gauss coefficients when finding magnetic-pole positions, so two pole positions cannot contain all information about Earth's field [3].

6 The South Atlantic Anomaly and Model Limits

The South Atlantic Anomaly is a useful example of where a simple dipole model breaks down. NASA describes it as a region of unusually low magnetic-field strength over South America and the South Atlantic [9]. Its position and shape also change with time.

A centered dipole cannot make a local weak-field region because its field is too symmetric. A displaced dipole can shift the overall field and create more uneven strength, so it is a better simple model. However, the weak point of a displaced-dipole calculation should not be called the true center of the SAA. A proper comparison needs a model such as WMM or IGRF that includes higher-order terms.

This is still useful. The centered and displaced models show the main geometry in a clear way. Their differences from the real SAA help show what the simple model is missing.

7 Scientific Computing and Visualization

The calculations were done in Wolfram Language. The notebook imports and cleans the two NOAA datasets, converts latitude and longitude to Cartesian coordinates, organizes the data by year, calculates $\boldsymbol{m}$, its magnitude, and its three direction angles, and repeats the process for all 436 years. It also calculates great-circle travel and creates several plots.

The main plots used in this report are the three direction-angle curves, the magnetic-moment magnitude, the three Cartesian components, cumulative pole travel, and the 3D pole paths. The notebook also creates a changing 3D view of the two poles and the vector between them. An animator can move through the years so that the secular variation can be seen instead of only read from numbers.

The hand calculations and computer work were used together. The 1697 and 2025 calculations were first done by hand. The notebook was then used to check those values and repeat the same process for every year in the dataset. This gave a way to verify the code before using it on the full dataset.

8 Discussion

8.1 What the simple model does well

The model does several useful things. It gives one clear magnetic axis from the two pole positions. It shows how that axis changes with time. It also shows that the north and south pole paths are not simple mirror images. Finally, the same vector can be placed into the standard dipole-field equation.

The project also connects several topics from vector calculus and physics. It uses coordinate conversion, vectors, dot products, angles, spherical distance, magnetic-field equations, numerical work, and 3D plots in one problem.

8.2 What the model does not do

The vector $\boldsymbol{m}$ in this project is not the true magnetic dipole moment of Earth. It is made by subtracting two unit-sphere position vectors, so its magnitude does not have the physical units of a real magnetic moment. Two pole positions are also not enough to rebuild the full magnetic field over Earth.

The model leaves out higher-order parts of the internal field, magnetic fields from Earth's crust, currents in the magnetosphere and ionosphere, and short-term magnetic disturbances. The NOAA pole track itself comes from a more complete field model. For that reason, this project should be viewed as a simple geometric model built from NOAA model outputs, not as a replacement for WMM or IGRF.

8.3 Meaning of the secular-variation results

The all-year plots show that there is no one constant drift rate from 1590 through 2025. The direction angles, model magnitude, and pole speeds all change over time. The 1697–2025 rate calculation is useful for comparing those two years, but it does not describe every year in the dataset.

The north pole also traveled farther than the south pole in this model. This supports the idea that the real geomagnetic field is more complicated than one centered dipole with two perfectly opposite endpoints.

9 Conclusions and Future Work

The NOAA pole data give a clear way to study secular variation with vector methods. By changing each pole position to Cartesian coordinates and defining $\boldsymbol{m} = \boldsymbol{r}_N - \boldsymbol{r}_S$, a model magnetic-moment vector can be found for every year from 1590 through 2025. The hand calculations for 1697 and 2025 agree with the computer results, which helped check the code before it was used on the whole dataset.

The main numerical results are:

- $\|\boldsymbol{m}_{1697}\| \approx 1.961898$ and $\|\boldsymbol{m}_{2025}\| \approx 1.930189$ arbitrary units;
- average 1697–2025 angle changes of about -1.3777°, 0.0928°, and 0.8629° per century for the x, y, and z axes;
- north-pole travel of about $0.7049R_E$, or about 4491 km;
- south-pole travel of about $0.4755R_E$, or about 3029 km;
- long-term average speeds of about 10.32 km/year for the north pole and 6.96 km/year for the south pole.

A centered dipole gives a useful first model of the field. A displaced dipole adds some asymmetry and is a reasonable next step. The South Atlantic Anomaly and the structure of WMM and IGRF show why these simple models are not enough for exact predictions.

Future work could include calculating centered and displaced surface-field maps for every year, comparing them directly with WMM or IGRF, measuring the error of each simple model, and then adding higher-order multipole terms. The project could also be extended to Science on a Sphere or to charged-particle motion in Earth's magnetic field.

10 Reflection and Acknowledgments

This project helped connect vector calculus to a real physical problem. The most important step for me was learning how to turn latitude and longitude into Cartesian vectors. Once that was done, the magnetic-moment vector, direction angles, great-circle distances, and 3D plots could all use the same coordinate system.

The project also showed me that a simple model can still be useful even when it is not exact. The centered dipole is easy to understand and calculate, but comparing it with the real Earth makes its limits clear. The displaced dipole improves part of the problem, but more terms are needed for a realistic magnetic field.

Dr. Héctor Mera proposed and supervised the independent-study project. He provided the MAT202 magnetic-field framework, discussed centered and displaced dipole models, and gave guidance on geomagnetism and Wolfram Mathematica. NOAA/NCEI supplied the magnetic-pole data and reference information for modern geomagnetic models. Wolfram Mathematica and Wolfram Cloud were used for calculations and plots. ChatGPT was used to help with Wolfram Language syntax and debugging, code organization, checking calculations, organizing research, and writing and editing the LaTeX report.

Reproducibility Note

The notebook contains the main computer steps used in the report. The process is: import and clean the NOAA data; match each year to a north and south pole position; use Eq. (1) to convert to Cartesian coordinates; calculate $\boldsymbol{m}$, its magnitude, and the direction angles; repeat for every year; calculate great-circle distances with Eq. (7); and create the plots. The hand work in Appendix A shows the same main calculations for 1697 and 2025.

A Hand Calculations

The following pages are copies of my original hand calculations. They show the coordinate-system setup, the latitude-longitude conversion, the 1697 and 2025 pole vectors, the model magnetic moments, the magnitudes, the direction angles, and the average angle-change rates. These pages were used as a check before the same calculations were automated in Wolfram Mathematica.

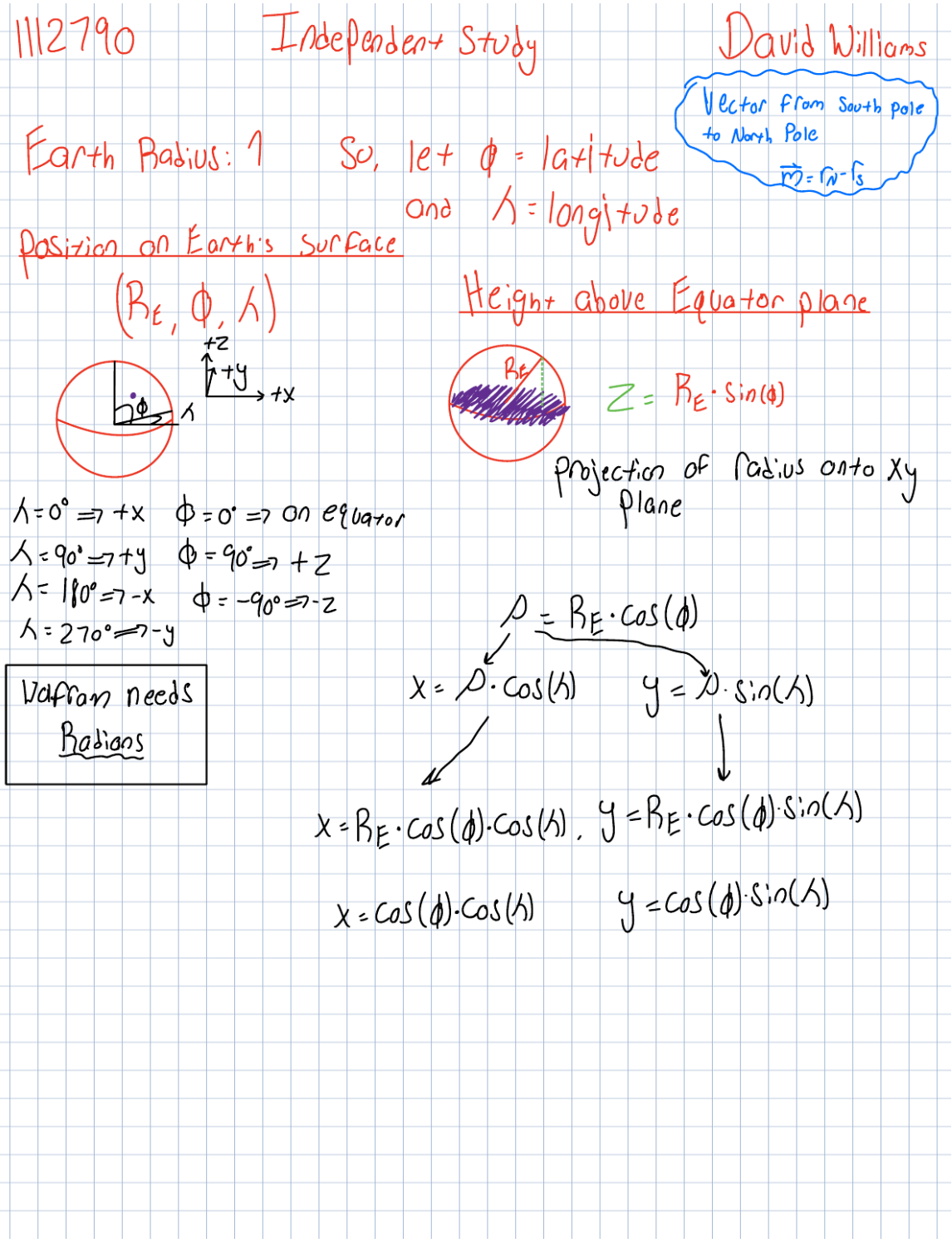


Figure 6: Hand derivation of the coordinate system and the latitude-longitude to Cartesian conversion.

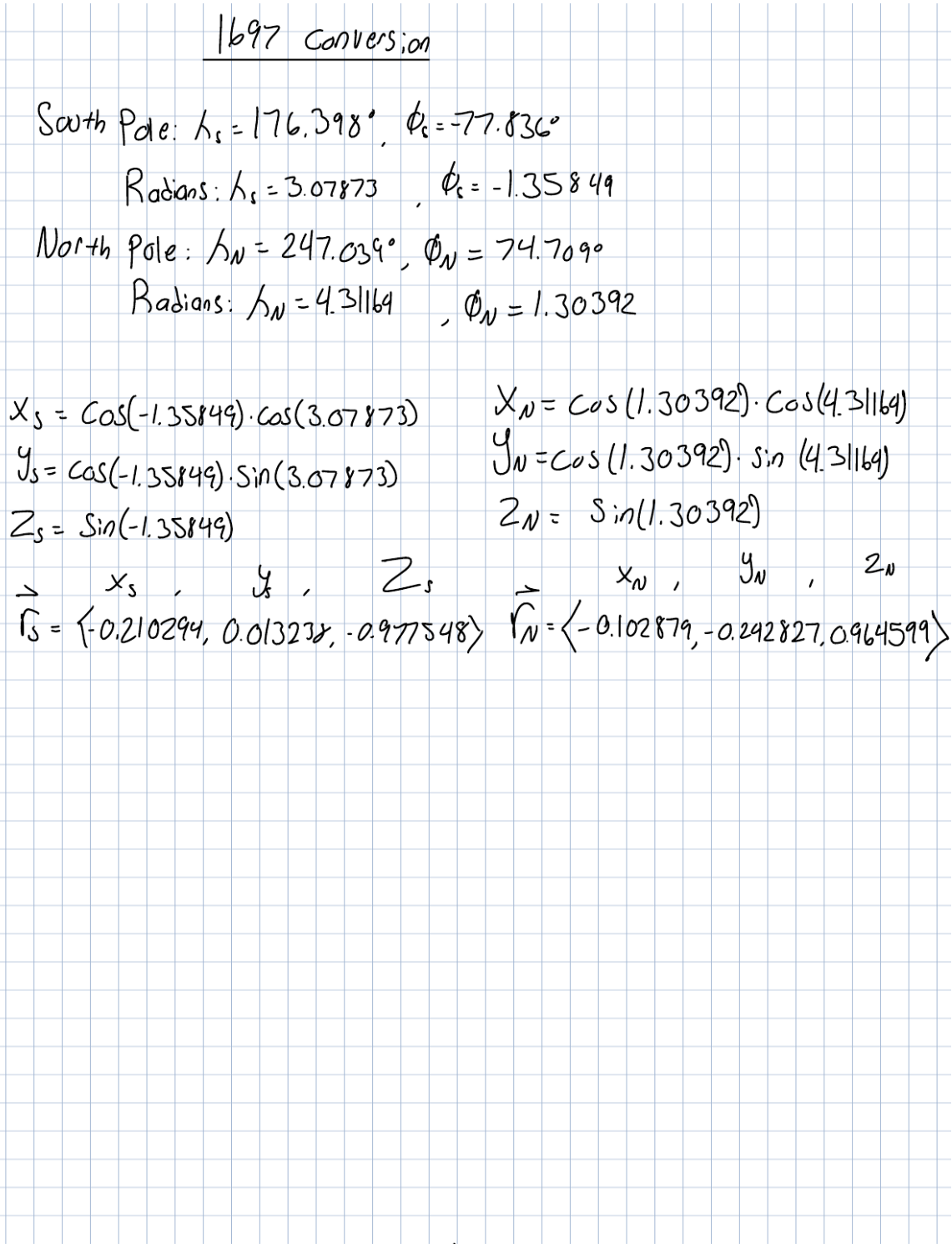


Figure 7: Hand conversion of the 1697 north and south magnetic-pole coordinates to Cartesian vectors.

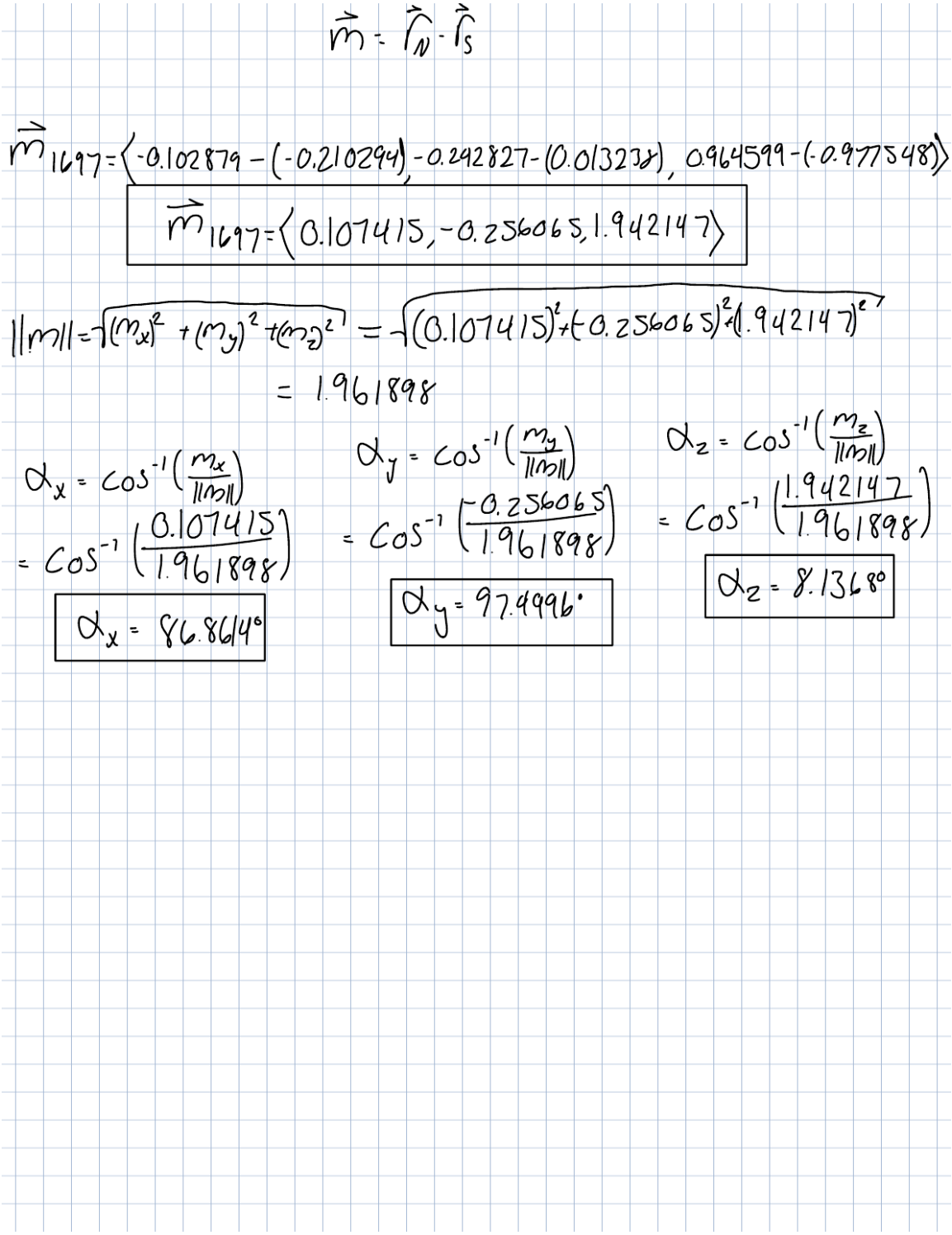


Figure 8: Hand calculation of the 1697 model magnetic moment, magnitude, and direction angles.

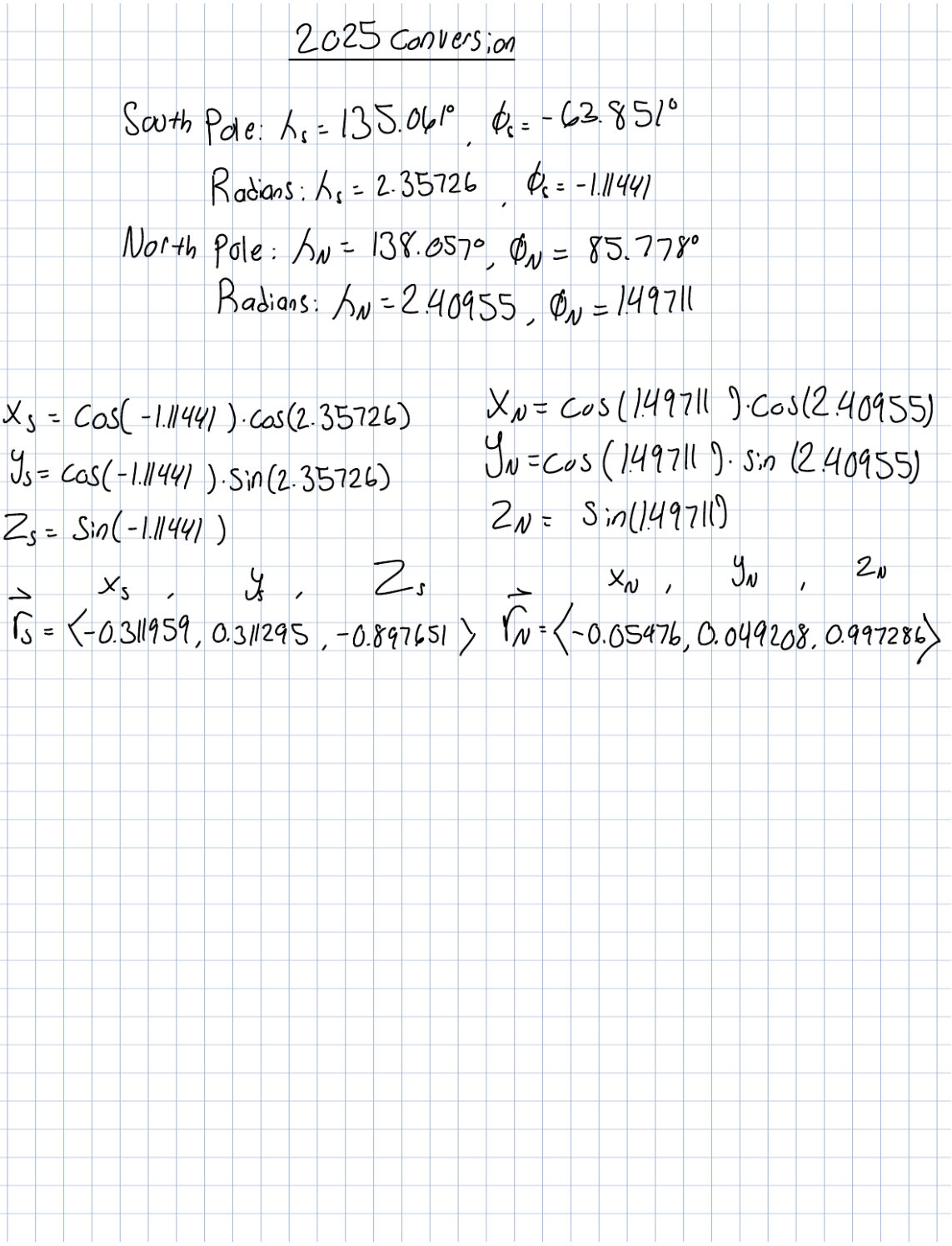


Figure 9: Hand conversion of the 2025 north and south magnetic-pole coordinates to Cartesian vectors.

$$\vec{m} = \vec{r}_N - \vec{r}_S$$

$$\vec{m}_{2025} = \langle -0.05476 - (-0.311959),\ 0.049208 - (0.311295),\ 0.997286 - (-0.897651) \rangle$$

$$\boxed{\vec{m}_{2025} = \langle 0.257198, -0.262087, 1.894937 \rangle}$$

$$\|m\| = \sqrt{(m_x)^2 + (m_y)^2 + (m_z)^2} = \sqrt{(0.257198)^2 + (-0.262087)^2 + (1.894937)^2}$$

$$= 1.93018$$

$$\alpha_x = \cos^{-1}\left(\frac{m_x}{\|m\|}\right) = \cos^{-1}\left(\frac{0.257198}{1.93018}\right)$$

$$\boxed{\alpha_x = 82.3425^\circ}$$

$$\alpha_y = \cos^{-1}\left(\frac{m_y}{\|m\|}\right) = \cos^{-1}\left(\frac{-0.262087}{1.93018}\right)$$

$$\boxed{\alpha_y = 97.8039^\circ}$$

$$\alpha_z = \cos^{-1}\left(\frac{m_z}{\|m\|}\right) = \cos^{-1}\left(\frac{1.894937}{1.93018}\right)$$

$$\boxed{\alpha_z = 10.9670^\circ}$$

Figure 10: Hand calculation of the 2025 model magnetic moment, magnitude, and direction angles.

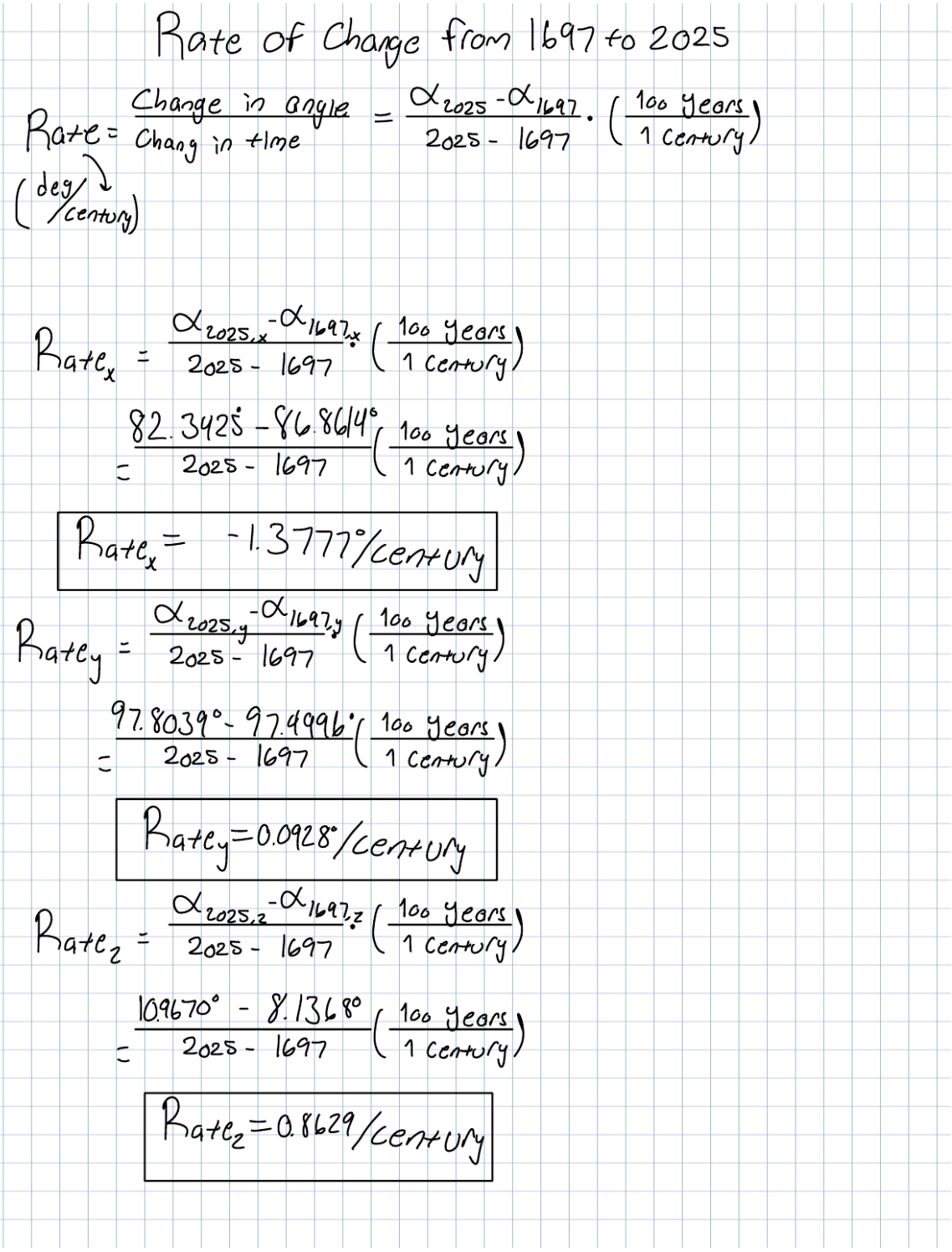


Figure 11: Hand calculation of the average angle-change rates from 1697 to 2025.

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